



HWA CHONG INSTITUTION
JC2 Preliminary Examinations
Higher 2

CANDIDATE NAME

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CT GROUP

20S7 ____

CENTRE NUMBER

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INDEX
NUMBER

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BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

15 September 2021

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

INSTRUCTIONS TO CANDIDATES

Write your **name**, **CT group**, **Centre number** and **index number** in the spaces at the top of this cover page.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

INFORMATION FOR CANDIDATES

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
1	/ 30
2	/ 10
3	/ 10
4 or 5	/ 25
Total	/ 75

This document consists of **20** printed pages.

QUESTION 1

A point mutation in the adenosine deaminase (*ADA*) gene results in a deficiency of ADA enzyme, which is a common cause of SCID.

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- [4]

SCID caused by ADA deficiency is a type of autosomal recessive disorder.

-
-
-
-
- [2]

Children with non-functional ADA are at a risk of infections as a toxic product builds up inside T lymphocytes and kills these cells.

- (c) (i)** With reference to **two** named T lymphocytes in the immune system, discuss how defective T lymphocytes increases the risk of infections.

[4]

- (ii)** Suggest why an increased risk of infections can be life threatening for individuals with SCID.

[1]

SCID can be treated using gene therapy with viral vectors, such as retroviruses. Viral vectors allow the therapeutic (normal) allele of a gene to be inserted into defective cells.

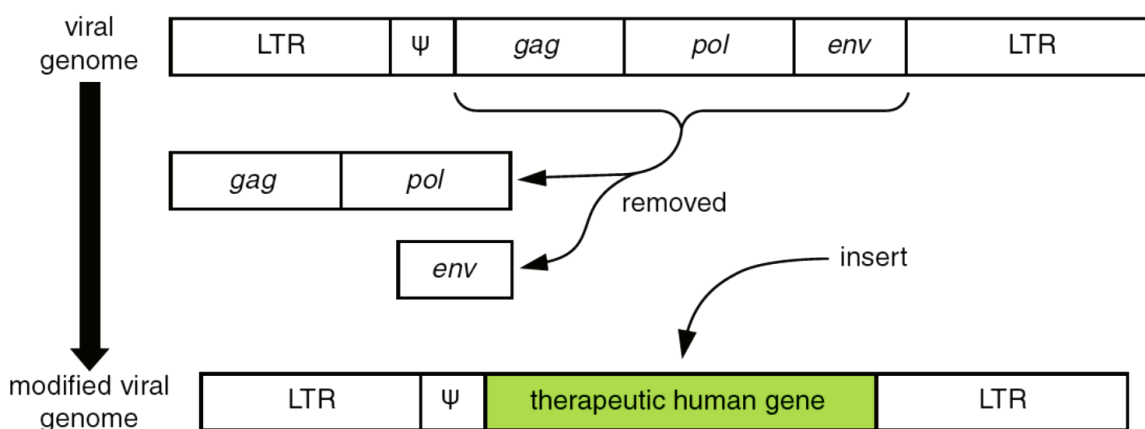
One method of gene therapy for SCID involves removing blood stem cells from an individual with SCID, modifying them using a viral vector and then putting the cells back into the individual.

- (d)** Suggest why stem cells are useful for gene therapy treatment.

[2]

Retroviruses have RNA as their genetic material. This is enclosed by a protein coat made of many protein subunits and an outer envelope made of a phospholipid bilayer.

The production of viral vectors from retroviruses for use in gene therapy firstly requires modification of the viral genome, summarised in Fig. 1.1.



key:

LTR = regulatory and control sequences such as enhancers

Ψ = sequence for directing the correct packaging of the genome into the protein coat during viral assembly

gag = codes for structural proteins for the protein coat

pol = codes for polyprotein, which is cleaved after synthesis to form two enzymes: integrase and reverse transcriptase

env = codes for viral proteins added to the outer envelope of the virus

Fig. 1.1

To obtain the viral vectors, modified mammalian cells known as packaging cells are used. These are engineered to contain in their nucleus the modified viral genome and the removed genes *gag*, *pol* and *env*. The purpose of a packaging cell is to replicate the viral genome and to synthesise viral proteins.

The sequence of events occurring in a packaging cell is:

- replication of the viral genome
- expression of *gag*, *pol* and *env* genes to produce viral proteins
- assembly of viral vector
- exit of viral vector.

The production of these viral vectors is summarised in Fig. 1.2.

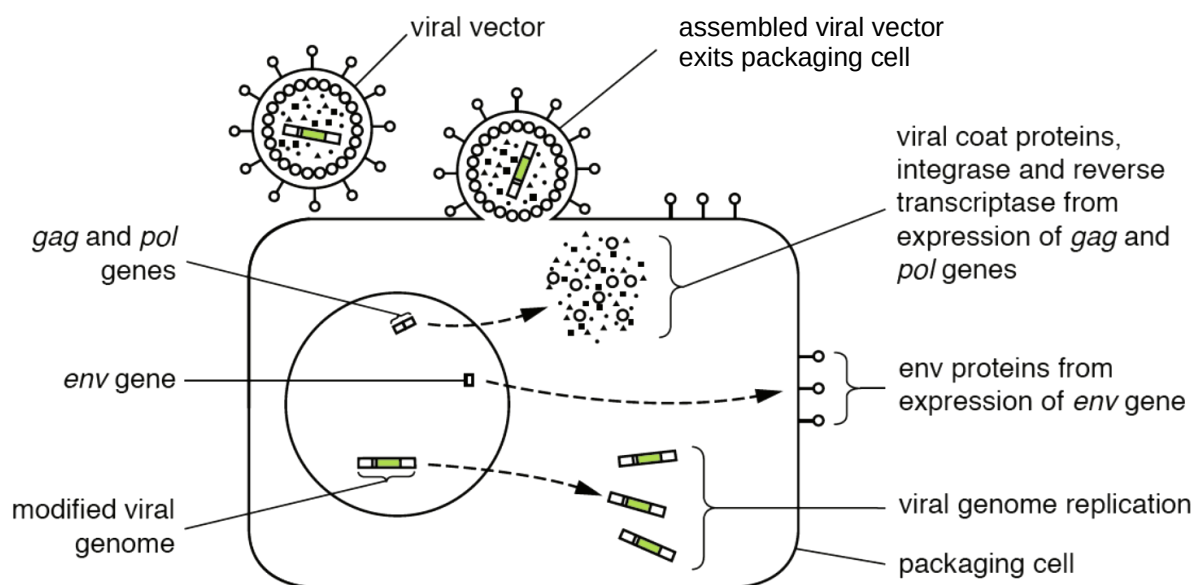


Fig. 1.2

Each viral vector produced contains integrase, reverse transcriptase, the modified viral genome and the viral envelope.

(e) With reference to Fig. 1.2,

- (i)** state the name of the enzyme that is used for the synthesis of RNA in viral genome replication.

[1]

- (ii)** describe how the viral vector obtains its viral envelope.

[3]

- (f) Suggest **and** explain the functions of the env proteins located in the viral envelope in the gene therapy treatment.

[2]

- (g) Following the successful uptake of the viral vector into the target cell,

- (i) state the role of reverse transcriptase.

[1]

- (ii) suggest possible roles of the LTR that is flanking the therapeutic human gene.

[2]

- (h) Suggest why it is important that the viral vector lacks the *gag*, *pol* and *env* genes as shown in Fig. 1.1.

[2]

The modified blood stem cells will differentiate into B and T lymphocytes, which are re-introduced into the individual with SCID.

To evaluate the level of expression of the therapeutic human gene in these cells, a technique called Real-Time Quantitative Reverse Transcription PCR (RT-qPCR) can be used.

Lymphocytes are first lysed. Their mRNA are then extracted and converted to single-stranded DNA (ssDNA) using a reverse transcriptase. The newly formed ssDNA is subsequently subjected to a modified PCR technique.

In addition to primers that flank the therapeutic gene and free nucleotides, an additional non-specific fluorescent dye that would bind to any double-stranded DNA (dsDNA) molecule are used for the PCR process. Fluorescence is detected when it crosses a threshold and the level of expression of the therapeutic human gene can be quantitatively determined.

The lymphocytes from two individuals, **A** and **B**, who had undergone gene therapy are extracted. The level of expression of the therapeutic gene are then quantified using RT-qPCR and the results are shown in Fig 1.3.

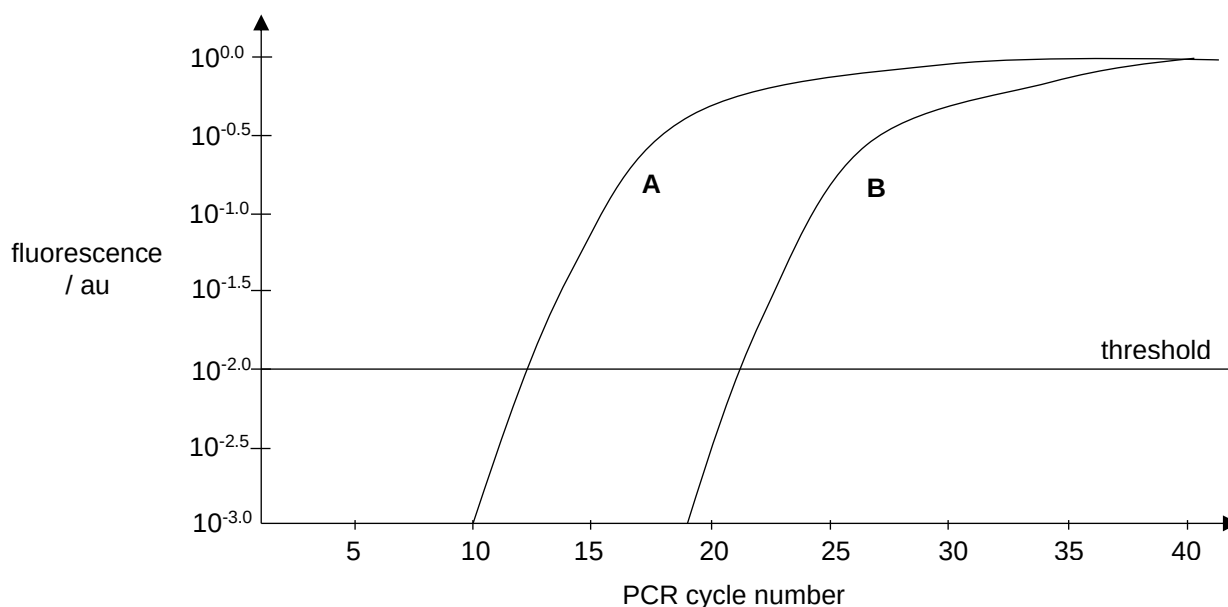


Fig. 1.3

- (i) Explain how the use of mRNA in RT-qPCR allows for quantification of the expression level of the therapeutic gene.

[2]

- (j) With reference to Fig 1.3, identify and explain which individual, **A** or **B**, would more likely to be successfully treated by gene therapy.

[3]

- (k) Suggest why individuals with SCID who have been successfully treated with gene therapy may still have offspring with SCID.

[1]

[Total: 30]

QUESTION 2

Corals are colonies of small animals (polyps) that grow in shallow sea water.

The coral reef ecosystem consists of corals that exist in symbiosis with a myriad of organisms such as:

- cyanobacteria, which are bacteria capable of both photosynthesis and nitrogen fixation.
- algae, which are protists capable of photosynthesis. The algae can live within the cells of coral polyps or live independently as free-living algae in the waters of the coral reef ecosystem.

This ecosystem is built upon an extensive carbonate skeleton of the corals, which serves as shelter and support for the cyanobacteria and algae.

Cyanobacteria and algae are both capable of using light to synthesise ATP. Fig 2.1 shows ATP synthesis within a chloroplast of an algal cell and in a cyanobacterium.

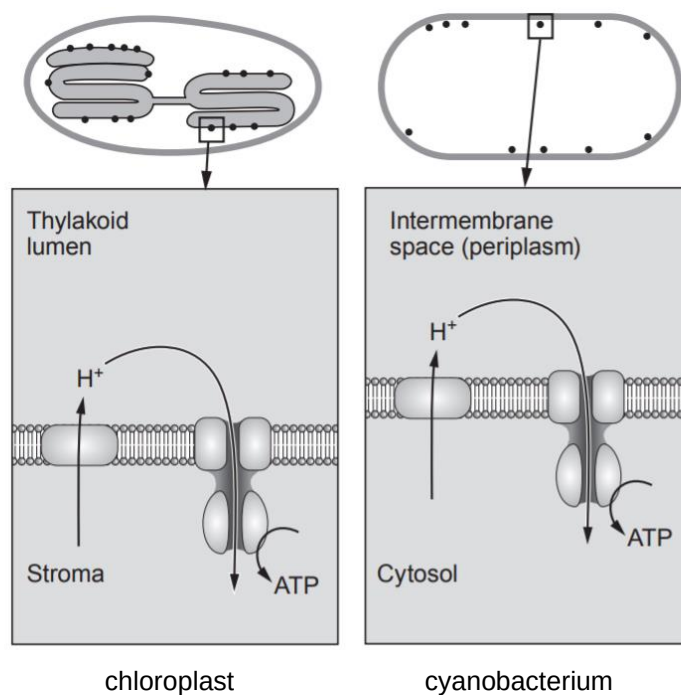


Fig. 2.1

- (a) State **two** similarities between ATP synthesis within a chloroplast of an algal cell and in a cyanobacterium.

[2]

- (b) The rate of photosynthesis of algae that live within the cells of coral polyps is higher than that of free-living algae.

Suggest **and** explain why the rate of photosynthesis in algae that live within the cells of coral polyps is higher than that of free-living algae.

[3]

The algae that live within the cells of coral polyps have five different chloroplast pigments.

Table 2.1 shows the light wavelengths at which each algal chloroplast pigment shows its two largest peaks of light absorption.

Table 2.1

chloroplast pigment	peak 1 wavelength / nm	peak 2 wavelength / nm
chlorophyll <i>a</i>	430	662
peridinin	456	485
chlorophyll <i>c</i> ₂	450	396
dinoxanthin	442	471
β-carotene	454	480

Corals can be kept in glass tanks that are usually lit by lamps radiating mainly violet and blue light, with wavelengths in the range of 400 nm to 490 nm.

- (c) With reference to Table 2.1, suggest why lamps radiating mainly violet and blue light are expected to increase the growth of coral polyps more than lamps radiating light of all wavelengths.

[3]

Despite its ability to photosynthesise, cyanobacteria are not the primary source of organic compounds for coral cells. Instead, cyanobacteria play a more important role in converting atmospheric nitrogen into a form that can be used by the corals.

(d) Explain how a reduced cyanobacteria population will result in death of corals.

[2]

[Total: 10]

QUESTION 3

The genomes of eukaryotes and prokaryotes differ greatly in size and number of protein-coding genes.

Fig. 3.1 shows the results of a study on eukaryotic and prokaryotic genomes as illustrated by using a boxplot. It is a visual representation comparing the distribution of datasets.

A boxplot consists of:

- the box that represents the data distribution through their quartiles, consisting of lower (25), median (50) and upper (75)
- the whiskers extending beyond the box that represent the two extreme quartiles, lower (0) and upper (100)
- the dots lying beyond the whiskers that represent data points deemed as outliers.

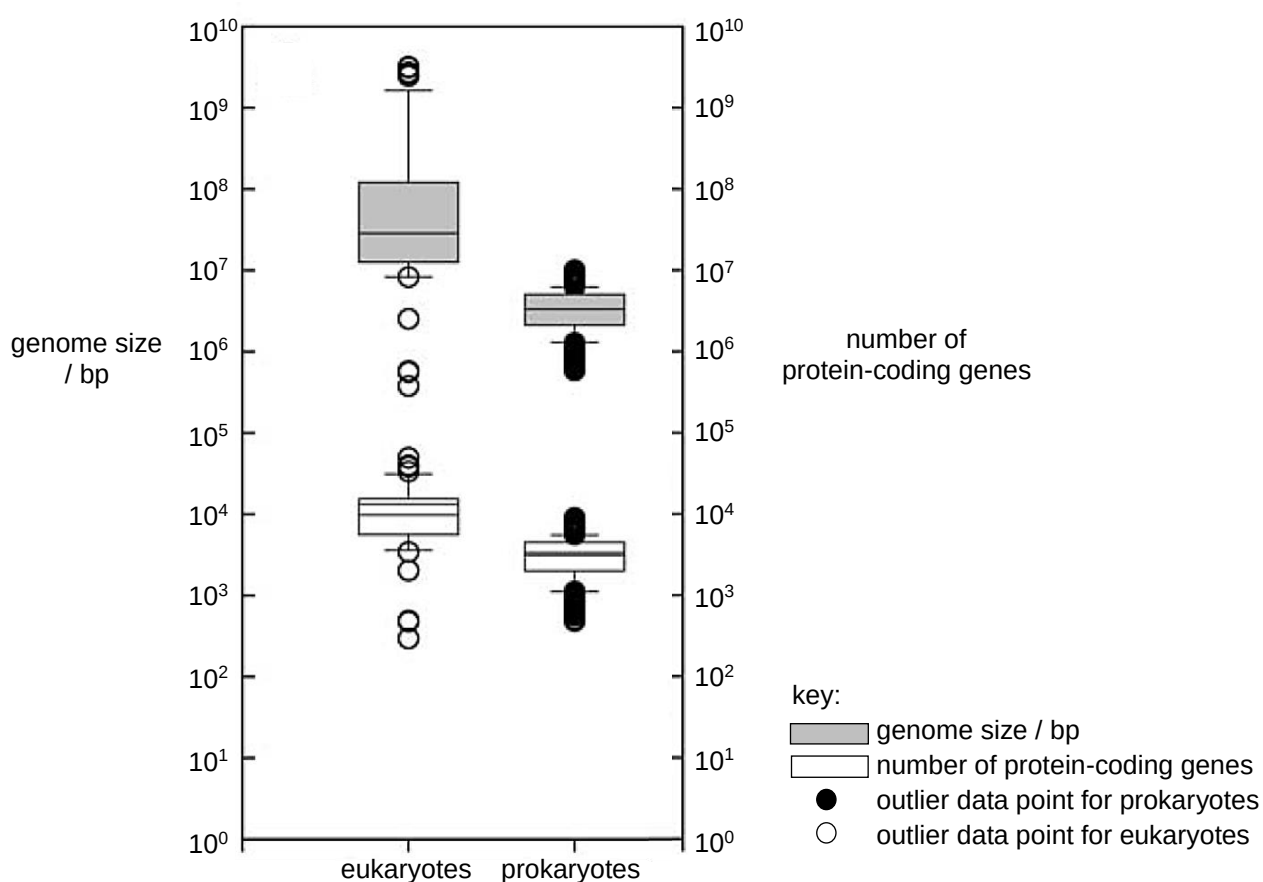


Fig. 3.1

(a) Describe **and** explain the results of the study shown in Fig 3.1.

[4]

(b) Suggest **two** advantages of using a boxplot.

[2]

The *cytochrome c* (*cyt c*) gene is highly conserved in eukaryotic genomes as cytochrome c is an essential protein for the process of oxidative phosphorylation.

A researcher wants to find out if the *cyt c* gene is also present in prokaryotes. He is provided with the following:

- a sample of eukaryotic DNA extract
- a sample of prokaryotic DNA extract
- fluorescent-labelled DNA probe complementary to part of the *cyt c* gene.

(c) Using your knowledge of molecular techniques, outline how the researcher can carry out this investigation.

[4]

[Total: 10]

--- END OF SECTION A ---

SECTION B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

QUESTION 4

- (a)** Discuss how variation in viral genomes arises and suggest how a bird influenza virus could infect a human and subsequently form a new strain of human influenza virus. [15]
- (b)** Prokaryotes differ from eukaryotes in terms of cell structure, genome organisation and cellular processes.

Outline the mechanism of asexual reproduction by binary fission in a typical bacterial cell and compare this with the mechanism of mitosis in a eukaryotic cell. [10]

[Total: 25]

QUESTION 5

- (a)** Among all nucleated body cells, only B and T lymphocytes lose DNA during their development and maturation.

Describe the processes where DNA is lost during the development of B lymphocytes before and after exposure to antigens and discuss the significance of this loss of DNA to the immune response. [15]

- (b)** Charles Darwin postulated that over long periods of time, descent with modification will eventually lead to the rich diversity of life.

Describe the mechanism for descent with modification and explain how variation in genetic information enables scientists to reconstruct phylogenies. [10]

[Total: 25]

--- END OF SECTION B ---

--- END OF PAPER---

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